

Final Comprehensive Examination – Medicinal Plants

Challenge / Very Hard Version (Lectures 5–12)

Time: 3 Hours

Total Marks: 100

Marking Scheme:

- 50 Questions × 2 marks = 100 marks

Instructions: - Answer all questions. - Choose the single best answer. - Questions integrate concepts across Lectures 5–12. - Difficulty level: **Very Hard / Top Students (A+)**.

Section A: Integrated Analytical MCQs (1–20)

Q1.

A leaf showing anomocytic stomata, striated cuticle, and beaded anticlinal walls most likely belongs to a plant used for:

A. Constipation B. Cardiac failure C. CNS disorders D. Urinary tract infections

Q2.

The laxative effect of Senna becomes clinically dangerous in intestinal obstruction because it:

A. Inhibits water absorption B. Increases peristalsis and luminal water simultaneously
C. Produces anticholinergic action D. Causes direct mucosal erosion

Q3.

Which combination BEST explains anticholinergic toxicity symptoms?

A. ↑ Acetylcholine – miosis – diarrhea B. ↓ Acetylcholine – mydriasis – dry skin
C. ↑ Histamine – bronchoconstriction D. ↓ Adrenaline – hypotension

Q4.

Hyoscine is preferred over atropine in motion sickness because it:

A. Has stronger cardiac action B. Produces greater CNS depression
C. Is free from anticholinergic effects D. Has shorter duration only

Q5.

A powdered drug giving a positive Vitali–Morin reaction should be avoided in patients with:

A. Constipation B. Glaucoma C. UTI D. Heart failure

Q6.

Which sequence correctly links plant → active constituent → therapeutic use?

A. Senna – saponins – expectorant B. Digitalis – cardenolides – cardiotonic

C. Uva Ursi – alkaloids – antispasmodic D. Ivy – anthraquinones – laxative

Q7.

Hydroquinone release from arbutin is clinically relevant because it:

A. Acts systemically as diuretic B. Produces antibacterial effect in alkaline urine

C. Stimulates bladder contraction D. Suppresses CNS

Q8.

Which feature is MOST useful to differentiate *Digitalis lanata* from *purpurea* microscopically?

A. Type of stomata B. Beaded anticlinal walls

C. Presence of lactone ring D. Number of sugars

Q9.

Rapid drying of *Digitalis* leaves is essential to:

A. Increase solubility B. Prevent enzymatic hydrolysis of glycosides

C. Improve color D. Enhance absorption

Q10.

Which combination indicates highest risk of cumulative toxicity?

A. Senna + hypokalemia B. Digitalis + narrow therapeutic index

C. Uva Ursi + tannins D. Ivy + saponins

Q11.

A patient presents with blurred vision, dry mouth, tachycardia, and hallucinations. The most likely offending plant is:

A. Uva Ursi B. Senna C. Datura D. Guava

Q12.

Which plant shows BOTH glandular and non-glandular trichomes with cardiotonic activity?

A. Belladonna B. Digitalis C. Ivy D. Senna

Q13.

Which statement correctly explains the prodrug nature of sennosides?

- A. They are activated in the stomach
- B. They are converted by colonic bacteria to active aglycones
- C. They act unchanged
- D. They require liver metabolism

Q14.

Which plant constituent is MOST responsible for greenish-brown urine discoloration?

- A. Sennosides
- B. Hydroquinone derivatives
- C. Cardiac glycosides
- D. Tropane alkaloids

Q15.

Which mechanism explains expectorant action of ivy?

- A. CNS depression
- B. Reflex gastric irritation by saponins
- C. Anticholinergic effect
- D. Direct bronchodilation only

Q16.

The presence of a five-membered lactone ring is diagnostic for:

- A. Tropane alkaloids
- B. Anthraquinones
- C. Cardenolides
- D. Flavonoids

Q17.

Which factor MOST contributes to Digitalis toxicity?

- A. Rapid elimination B. Accumulation in body tissues
- C. Low potency D. Wide safety margin

Q18.

Which plant is LEAST suitable for long-term use?

- A. Guava B. Uva Ursi C. Ivy D. Tea

Q19.

Which stomatal type lacks subsidiary cells completely?

- A. Diacytic B. Paracytic C. Anisocytic D. Anomocytic

Q20.

Leaf anatomy is MOST important in pharmacognosy for:

- A. Increasing yield B. Correct identification of crude drugs
- C. Improving taste D. Determining dose

Section B: Case-Based & Cross-Link MCQs (21–50)

Q21.

A cardiac patient on Digitalis develops arrhythmia after prolonged use. This is MOST likely due to:

- A. Hypocalcemia B. Drug accumulation C. Rapid metabolism D. Reduced absorption

Q22.

A patient with chronic constipation and electrolyte imbalance should avoid:

- A. Guava B. Senna C. Ivy D. Tea

Q23.

Which plant acts by blocking muscarinic receptors but is used therapeutically in low doses?

- A. Uva Ursi B. Belladonna C. Digitalis D. Guava

Q24.

A drug acting mainly in the colon and not absorbed in stomach is:

A. Senna B. Digitalis C. Hyoscine D. Eucalyptus

Q25.

Which feature confirms Solanaceae plants belong to same family?

A. Habitat B. Flower color C. Active constituents and pharmacological action D. Leaf size

Q26.

Which plant should be avoided in glaucoma?

A. Ivy B. Uva Ursi C. Datura D. Guava

Q27.

The stimulant effect of tea is BEST explained by:

A. Tannins B. Flavonoids C. Caffeine D. Saponins

Q28.

Which plant combines astringent and antibacterial effects?

A. Senna B. Guava C. Digitalis D. Ivy

Q29.

Which condition contraindicates Uva Ursi long-term use?

A. Renal stones B. Gastric irritation C. Hypertension D. Asthma

Q30.

Which structural feature distinguishes cardiac glycosides from other glycosides?

A. Aromatic nucleus B. Steroidal nucleus with lactone ring C. Indole nucleus D. Purine ring

Q31.

Which plant is most appropriate for productive cough with thick sputum?

A. Guava B. Ivy C. Senna D. Tea

Q32.

Which examination technique best differentiates *Digitalis* species?

A. Organoleptic B. Microscopic C. Chemical only D. Taste evaluation

Q33.

A patient shows anticholinergic crisis (HORSE MAD). The causative plant likely contains:

A. Alkaloids of tropane type B. Anthraquinones C. Cardenolides D. Saponins

Q34.

Which factor makes *Digitalis* a high-risk drug?

A. Poor absorption B. Narrow therapeutic index C. Low potency D. Rapid clearance

Q35.

Which plant's action depends on urine alkalinity?

A. Senna B. *Digitalis* C. *Uva Ursi* D. Ivy

Q36.

Which pair is correctly matched?

A. Senna – anticholinergic B. Belladonna – parasympathomimetic

C. *Digitalis* – cardiotonic D. Ivy – laxative

Q37.

Which plant is contraindicated in intestinal obstruction?

A. Guava B. Senna C. Tea D. Ivy

Q38.

Which plant contains acetylated glycosides?

A. *Digitalis lanata* B. *Digitalis purpurea* C. Senna D. Guava

Q39.

Which microscopic feature is LEAST helpful in identification?

A. Stomatal type B. Anticlinal wall pattern C. Leaf taste D. Trichomes

Q40.

Which drug acts mainly by increasing myocardial contractility?

A. Senna B. Digitalis C. Uva Ursi D. Belladonna

Q41.

Which plant shows reflex expectorant action?

A. Guava B. Ivy C. Tea D. Digitalis

Q42.

Which constituent is responsible for hypokalemia risk?

A. Sennosides B. Tannins C. Caffeine D. Saponins

Q43.

Which plant drug should be avoided in elderly cardiac patients?

A. Digitalis (without monitoring) B. Guava C. Ivy D. Tea

Q44.

Which stomatal arrangement shows subsidiary cells parallel to guard cells?

A. Diacytic B. Anomocytic C. Paracytic D. Anisocytic

Q45.

Which plant acts mainly on CNS vestibular pathways?

A. Hyoscine B. Senna C. Uva Ursi D. Ivy

Q46.

Which feature explains cumulative Digitalis toxicity?

A. High water solubility B. Tissue binding C. Rapid metabolism D. Low potency

Q47.

Which plant should be avoided in chronic use due to hepatotoxicity risk?

A. Guava B. Uva Ursi C. Ivy D. Tea

Q48.

Which plant's efficacy depends on bacterial metabolism?

A. Senna B. Digitalis C. Guava D. Ivy

Q49.

Which pharmacological action links Belladonna, Datura, and Hyoscyamus?

A. Cardiotonic B. Anticholinergic C. Expectorant D. Diuretic

Q50.

Which statement BEST describes a Challenge-level exam question?

A. Recall-based B. Definition-based

C. Concept integration across topics D. Memorization only

End of Exam

Model Answers & Explanations

Final Comprehensive Exam – Medicinal Plants (Lectures 5–12)

Purpose: Instructor reference & final student review

Answer Key + Explanations (Challenge Version)

Q1. **C**

Anomocytic stomata and striated cuticle are characteristic microscopic features of Solanaceae plants acting on CNS.

Q2. **B**

Senna increases peristalsis and water secretion, which is dangerous in obstruction.

Q3. **B**

Anticholinergic toxicity results from blocking acetylcholine causing dry skin, mydriasis, and tachycardia.

Q4. **B**

Hyoscine causes greater CNS sedation, making it suitable for motion sickness.

Q5. **B**

Tropane alkaloids increase intraocular pressure and worsen glaucoma.

Q6. **B**

Digitalis contains cardenolides responsible for cardiotonic action.

Q7. **B**

Hydroquinone released from arbutin exerts antibacterial effect in alkaline urine.

Q8. **B**

Beaded anticlinal walls are characteristic of *Digitalis lanata*.

Q9. **B**

Rapid drying prevents enzymatic hydrolysis of cardiac glycosides.

Q10. B

Digitalis has a narrow therapeutic index leading to toxicity.

Q11. C

Datura causes classic anticholinergic toxicity symptoms.

Q12. B

Digitalis shows both glandular and non-glandular trichomes.

Q13. B

Sennosides are activated by colonic bacteria.

Q14. B

Hydroquinone oxidation causes greenish-brown urine discoloration.

Q15. B

Saponins act reflexly as expectorants via gastric irritation.

Q16. C

Cardenolides possess a five-membered lactone ring.

Q17. B

Digitalis accumulates in tissues causing toxicity.

Q18. B

Long-term use of Uva Ursi is avoided due to tannins and hepatotoxicity risk.

Q19. D

Anomocytic stomata lack subsidiary cells.

Q20. B

Leaf anatomy ensures correct identification of crude drugs.

Q21. B

Digitalis toxicity occurs due to accumulation.

Q22. B

Senna causes electrolyte imbalance.

Q23. B

Belladonna blocks muscarinic receptors therapeutically in low doses.

Q24. A

Senna acts mainly in the colon.

Q25. C

Shared active constituents explain same family classification.

Q26. C

Anticholinergics worsen glaucoma.

Q27. C

Caffeine is responsible for tea's stimulant effect.

Q28. B

Guava combines astringent and antibacterial actions.

Q29. B

High tannin content causes gastric irritation.

Q30. B

Steroidal nucleus with lactone ring defines cardiac glycosides.

Q31. B

Ivy saponins are effective in productive cough.

Q32. B

Microscopy differentiates Digitalis species.

Q33. A

Tropane alkaloids cause anticholinergic crisis.

Q34. B

Narrow therapeutic index makes Digitalis high-risk.

Q35. C

Uva Ursi requires alkaline urine.

Q36. C

Digitalis is cardiotonic.

Q37. B

Senna is contraindicated in obstruction.

Q38. A

Lanata glycosides contain acetyl groups.

Q39. C

Taste is least reliable microscopically.

Q40. B

Digitalis increases myocardial contractility.

Q41. B

Ivy acts reflexly as expectorant.

Q42. A

Sennosides cause potassium loss.

Q43. A

Unmonitored Digitalis is dangerous in elderly.

Q44. C

Paracytic stomata have parallel subsidiary cells.

Q45. A

Hyoscine acts on vestibular pathways.

Q46. B

Tissue binding leads to accumulation.

Q47. B

Uva Ursi long-term use risks hepatotoxicity.

Q48. A

Senna requires bacterial metabolism.

Q49. B

All are anticholinergic Solanaceae plants.

Q50. C

Challenge questions integrate concepts across topics.

End of Model Answers & Explanations